

## Non-Functional Requirements Document — Application

### Performance Requirements

- System must process and load customer and order data from source CSV files into data tables within 30 minutes for data volumes up to 1 million records.
- Aggregation and data update operations (including SCD Type 2 history updates) must complete within 20 minutes per batch.
- Average latency for record insertion or modification must not exceed 2 seconds for single-record operations.
- Concurrency support for at least 10 simultaneous processing jobs.
- Throughput of 10,000 records/minute for ingestion jobs.

### Scalability Requirements

- System must support horizontal scaling for increasing data volumes up to 10 million records.
- Must be able to add compute resources dynamically (vertical scaling) as needed during high-load periods.
- Architecture should support clustering for distributed data processing.
- System must accommodate future schema extensions with minimal downtime.

### Availability Requirements

- System must provide 99.5% uptime (excluding scheduled maintenance).
- Redundant data storage must be utilized to prevent single points of failure.
- Failover mechanisms must be in place to handle hardware or software failures without data loss.

### Reliability & Resilience Requirements

- Mean Time To Recovery (MTTR) must not exceed 1 hour for processed data.
- Mean Time Between Failures (MTBF) must be at least 1,000 hours of operation.
- Incremental backup and point-in-time restore capabilities must be implemented.
- Robust data validation and integrity checks are required at every transformation step.

### Security Requirements

- User access controls must be enforced for all Delta tables and transformation jobs.
- Role-based access must be implemented for data and jobs.
- All data at rest and in transit must be encrypted using industry best-practice standards (e.g., AES-256).
- Detailed audit logging of all data loads, updates, and deletions must be maintained.
- Data masking or tokenization must be applied to sensitive fields in non-production environments.

### Compliance Requirements

- Solution must comply with GDPR (where applicable), including right to erasure and data minimization.
- Adhere to SOX and other applicable local data governance standards.
- Ensure audit trails for all data processing pipelines.
- Data retention and archival policies must be clearly defined and followed.

#### Maintainability Requirements

- Source code for all processing jobs must be under version control.
- Clear coding standards and process documentation must be maintained.
- Automated CI/CD pipelines for deployment, validation, and rollback must be established.
- System configuration must be externalized and parameter-driven.
- Documentation for all batch jobs, tables, and column definitions is required.

#### Observability Requirements

- All job executions, successes, and failures must be logged and traceable.
- Real-time dashboards for job status, performance metrics, and data quality must be available.
- Integrate with enterprise monitoring solutions for alerting and historical metrics.
- System must record trace events for debugging and audit purposes.

#### Usability & Accessibility Requirements

- Any user-accessible dashboards or interfaces must conform with WCAG 2.1 Level AA accessibility guidelines.
- System status and job outcomes must be presented with clear notifications and error messages.
- Documentation and guides must be accessible to all approved stakeholders.

#### Service Level Agreements (SLAs) & Service Boundaries

- Data ingestion and processing must meet the performance and uptime SLAs stated.
- System interfaces must document input/output contracts and expected behaviors.
- Service boundaries with upstream and downstream systems must be documented, including data schemas and failure handling.

#### Constraints and Dependencies

- Performance dependent on source storage and delta processing cluster capacity.
- Compliance with enterprise IAM, logging, and encryption policies required.
- Minimum supported runtime version(s) must be documented.
- Adherence to corporate backup, archival, and data purging policies.

## Engineering Considerations

- Solution must follow ISO/IEC 25010 quality attribute guidelines for reliability, maintainability, and security.
- Reusable, modular job code is required to support maintainability and future extensibility.
- All platform dependencies must be identified and documented.

## References

- ISO/IEC 25010:2011 Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)
- GDPR, SOX regulatory texts
- WCAG 2.1 guidelines

## Definitions

- MTTR: Mean Time to Recovery
- MTBF: Mean Time Between Failures
- SCD: Slowly Changing Dimension
- SLA: Service Level Agreement
- IAM: Identity and Access Management
- CI/CD: Continuous Integration/Continuous Deployment